

Course Reminders

- Final Project due Wed, Mar 19th (11:59 PM)
 - Report (GitHub)
 - Video (one person for group submits on Canvas)
 - Team Evaluation Survey: [link to survey](#) (link also on Canvas; required)
- My office hours today are 10-11:20
- Post COGS 108 Survey: [link to survey](#) (link also on Canvas; *optional* for EC)
- SETs (Complete by Sat 3/15 by 8:00AM)
- Please do not email me about rounding your grade up

Jobs and the State of Data Science

Shannon E. Ellis, Ph.D
UC San Diego



Department of Cognitive Science
sellis@ucsd.edu

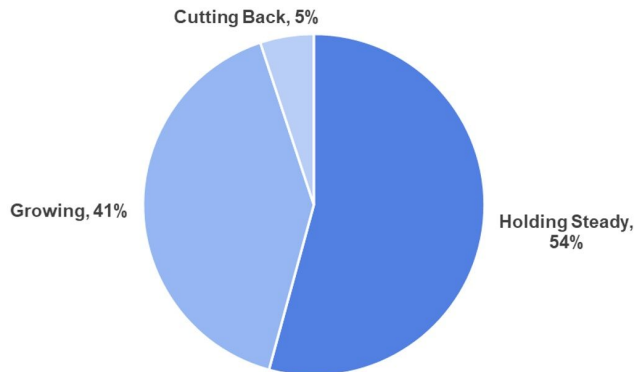
Getting a sense for where we're all at
(complete all questions; there are no right answers)



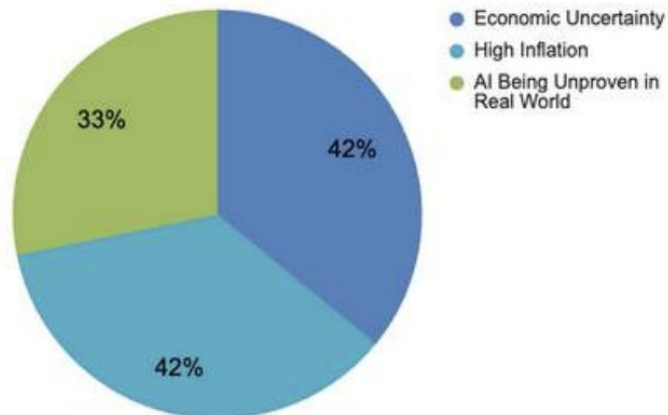
Burtchworks annual predictions and report on DS hiring

“Hiring is Stabilizing, but Growth Remains Selective”

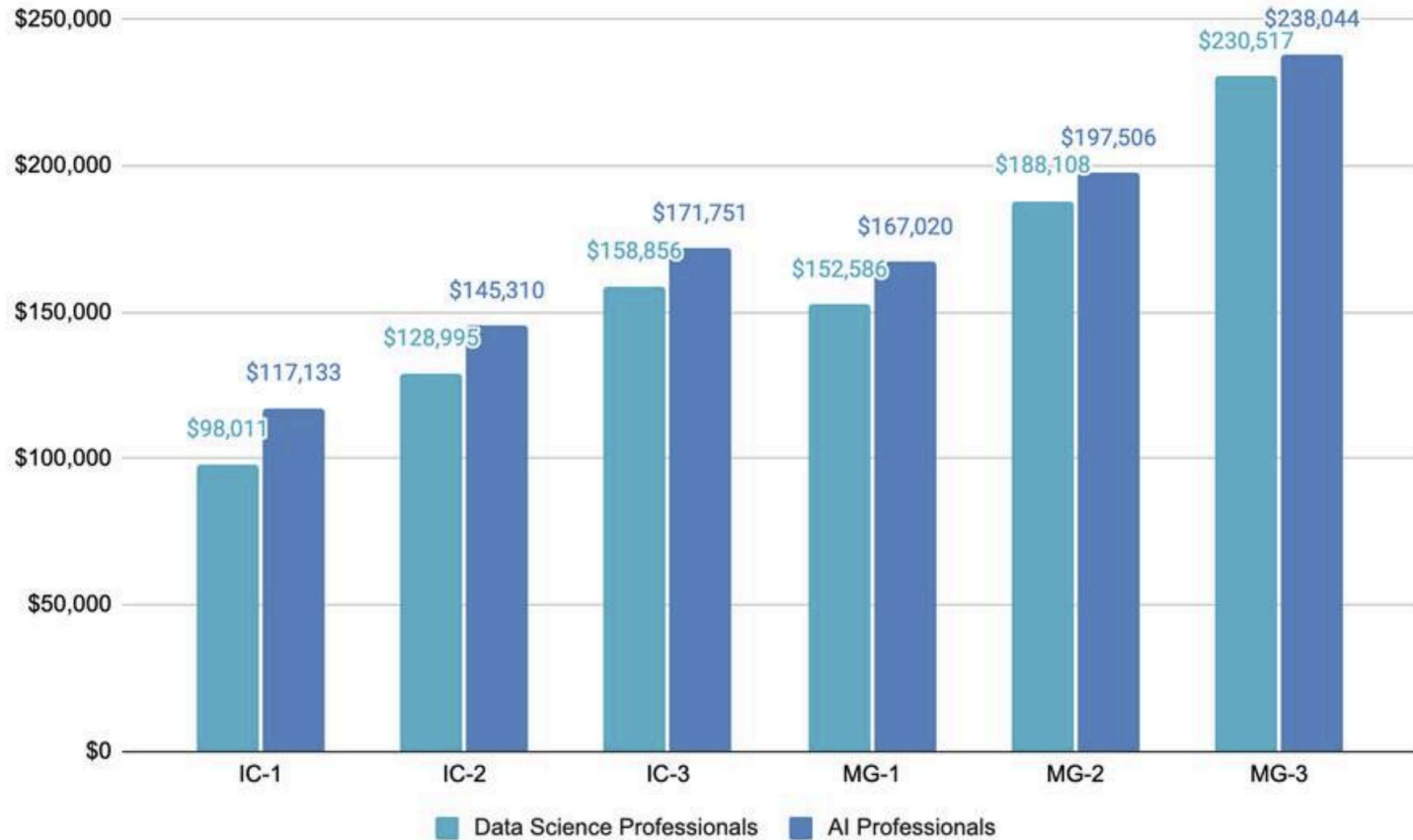
Hiring Plans



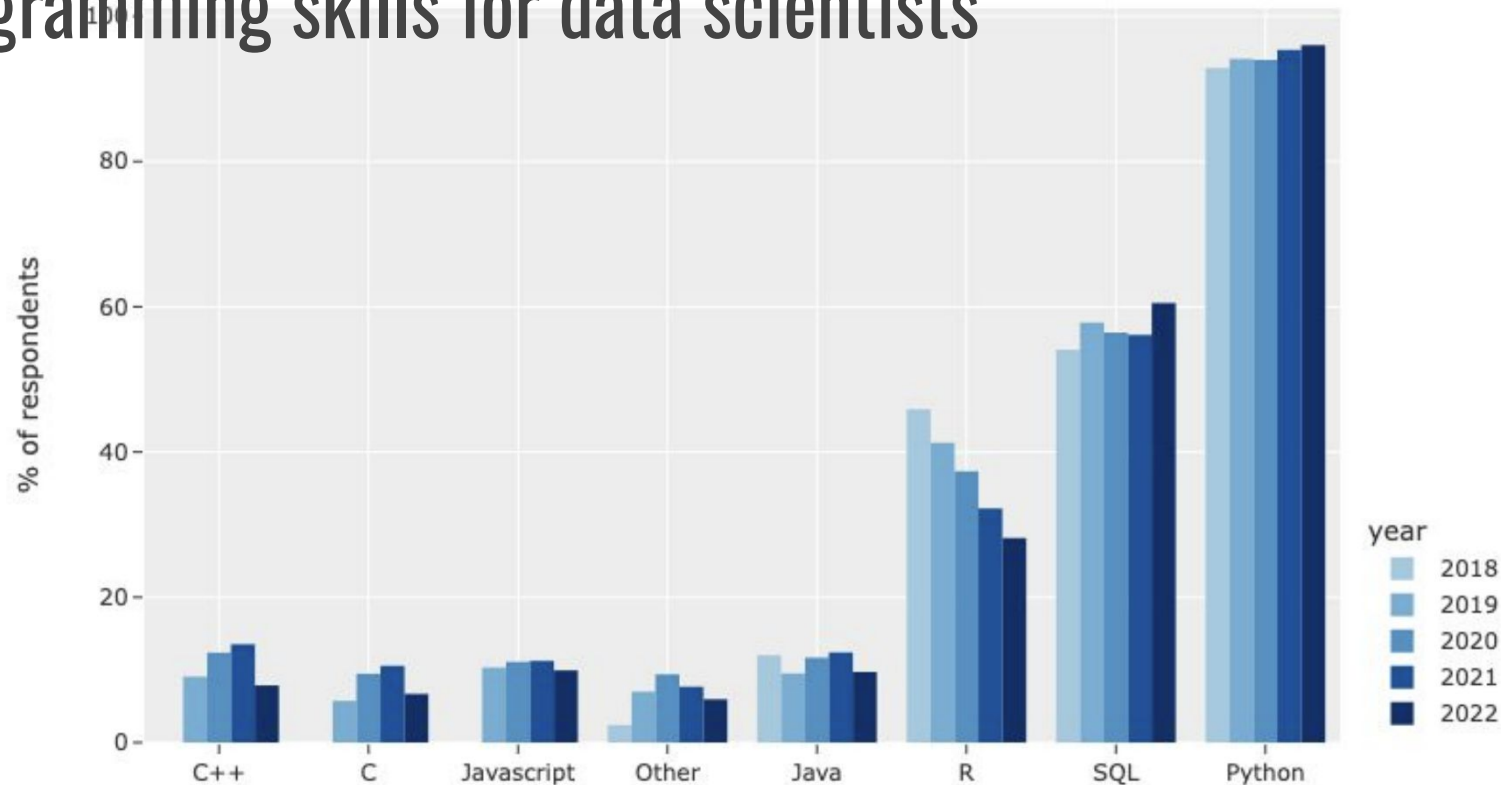
Economic conditions impacting hiring needs the most:

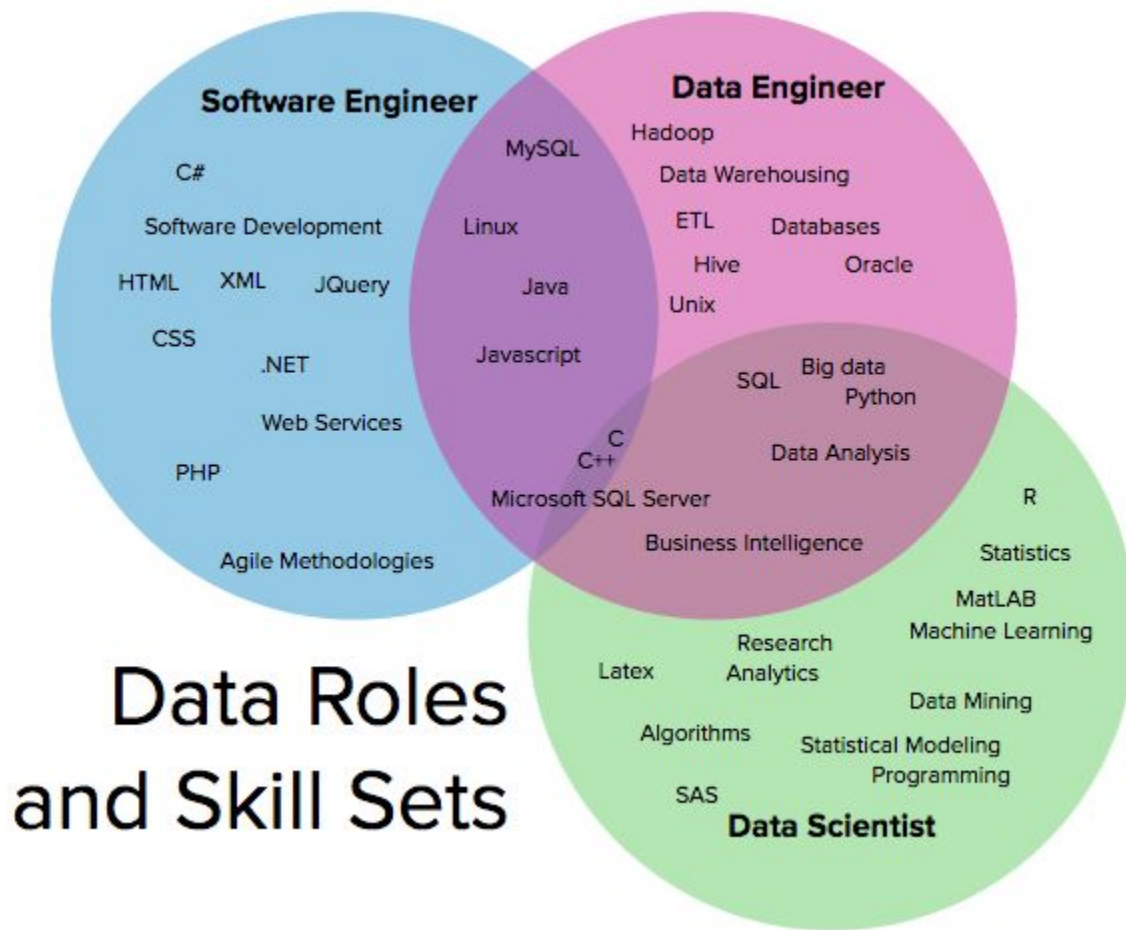


2024 Mean Base Salaries

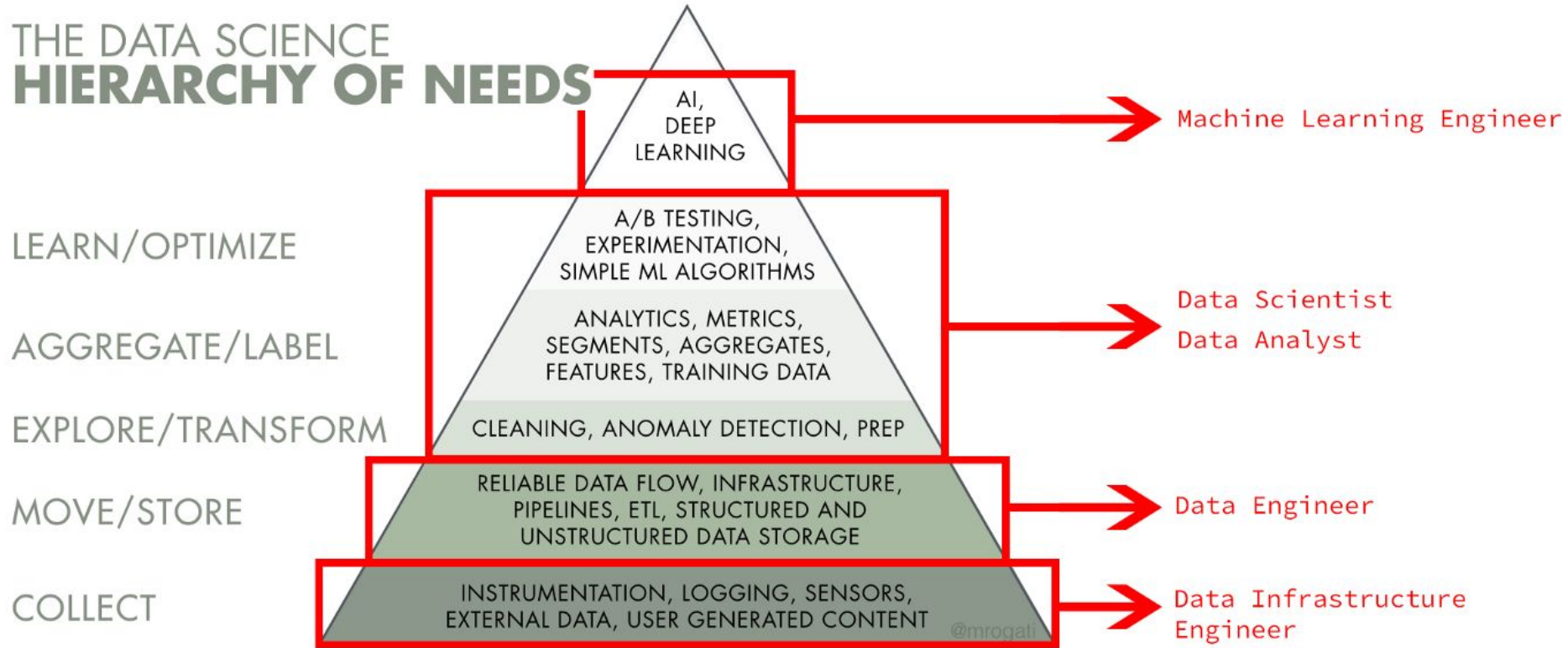


Python & SQL are consistently the two most common programming skills for data scientists





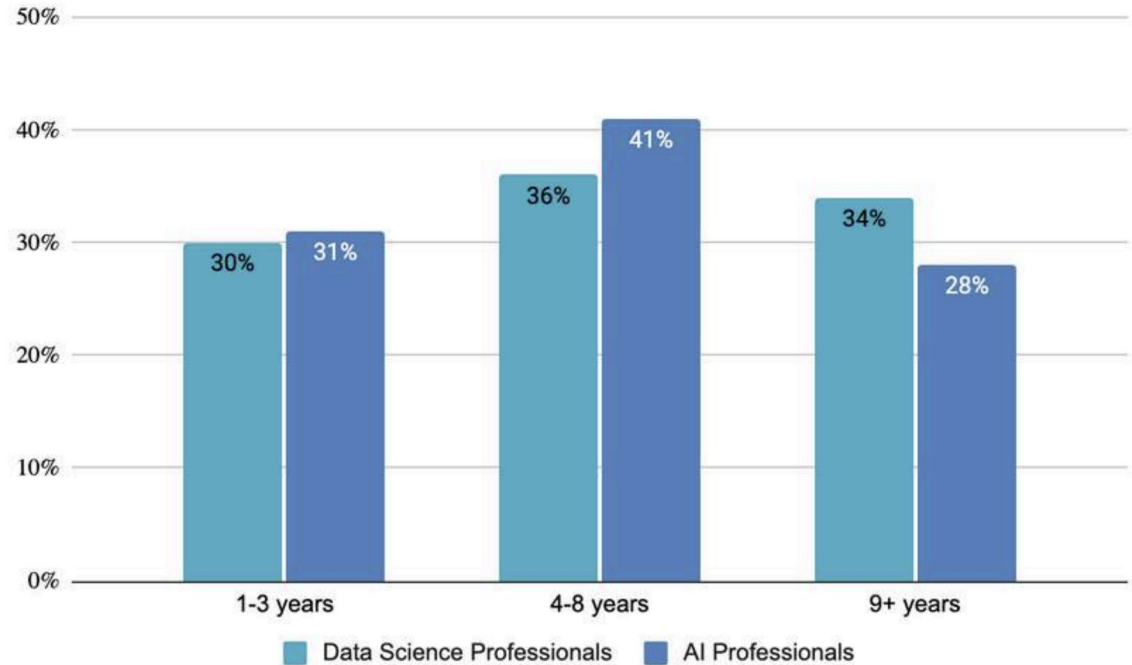
THE DATA SCIENCE HIERARCHY OF NEEDS



Some DS-related job titles

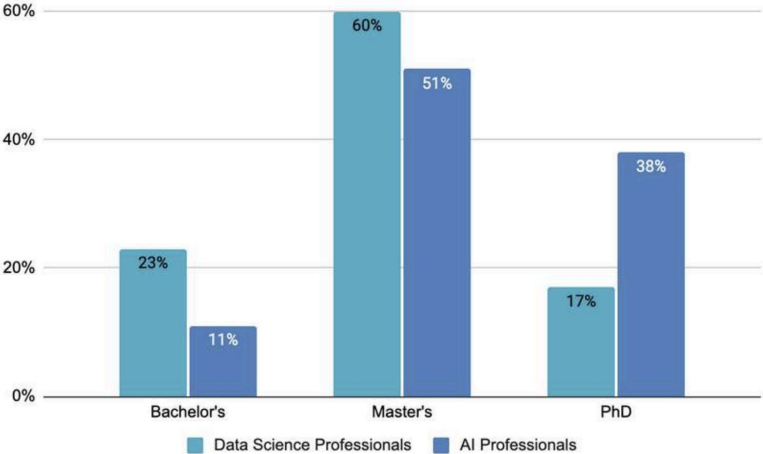
- Analytics or statistician: data handling, analysis
- Data Journalist: programming, viz, communication
- Data analyst: programming, data handling, reporting/metric tracking, viz
- Data scientist: programming, data handling, analysis
- Data engineer: programming, databases, management
- Data architect: programming, databases, design
- Data manager: databases, design, management
- *OPs (eg, devOPs, dataOPs, full stack): programming, tool development, management
concentrating on end to end process
- ML Engineer: programming, tool development, management of infrastructure
- ML researcher: programming, algorithm design and testing

Distribution of Data Science and AI Professionals by Years of Experience

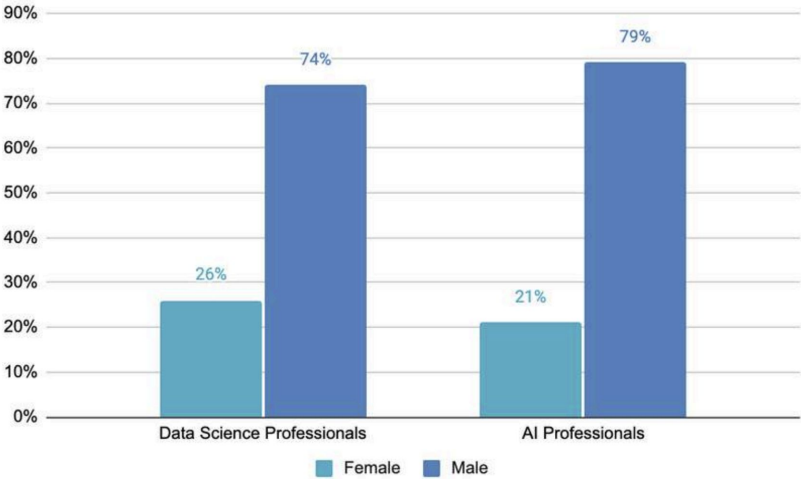


Masters are most common degree

COMPARISON OF DEGREE LEVEL *(for highest degree earned)*
Data Science Professionals vs. AI Professionals

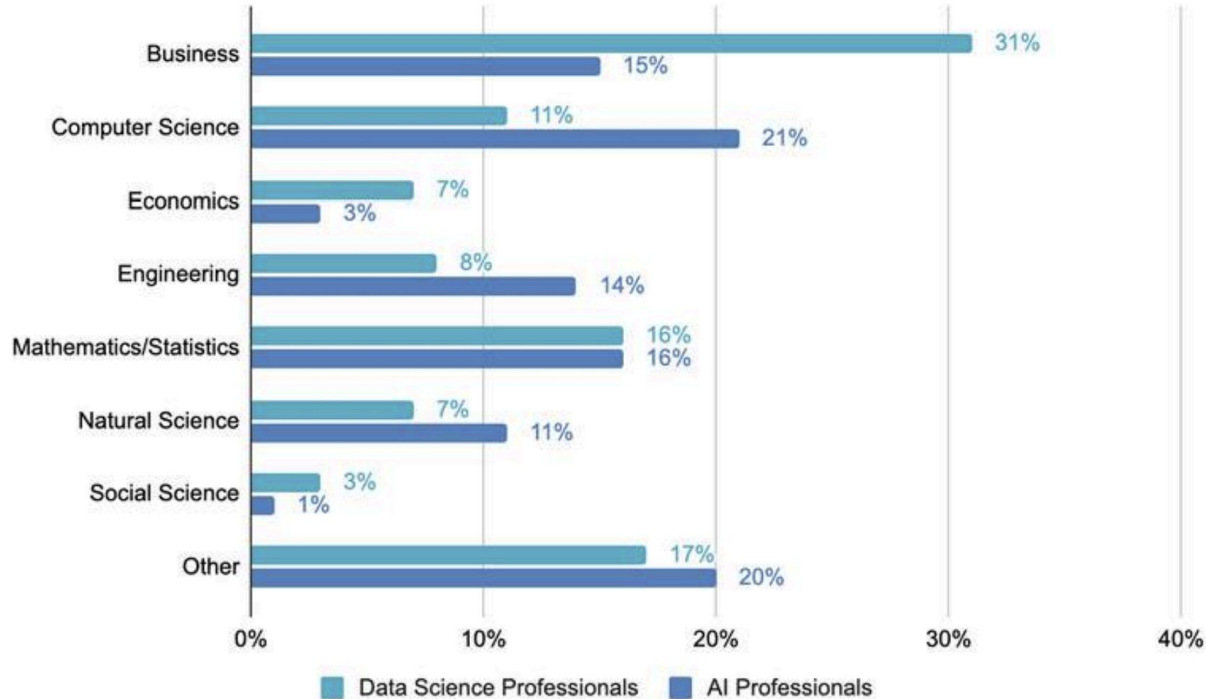


Consistent gender imbalance across the field

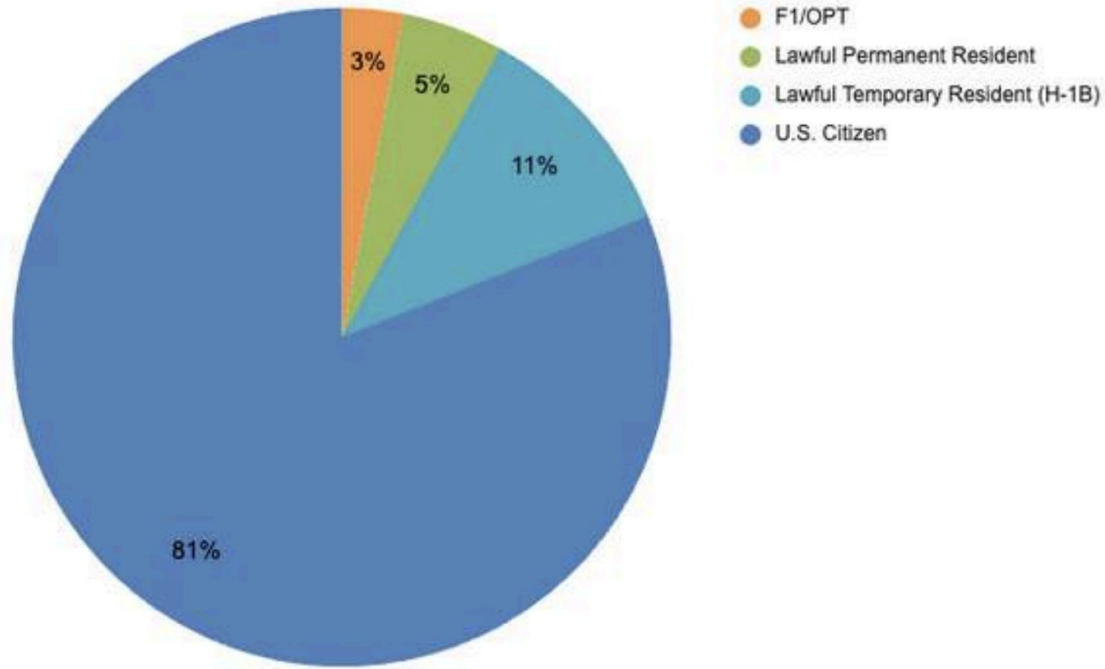


COMPARISON OF AREA OF report (for highest degree earned)

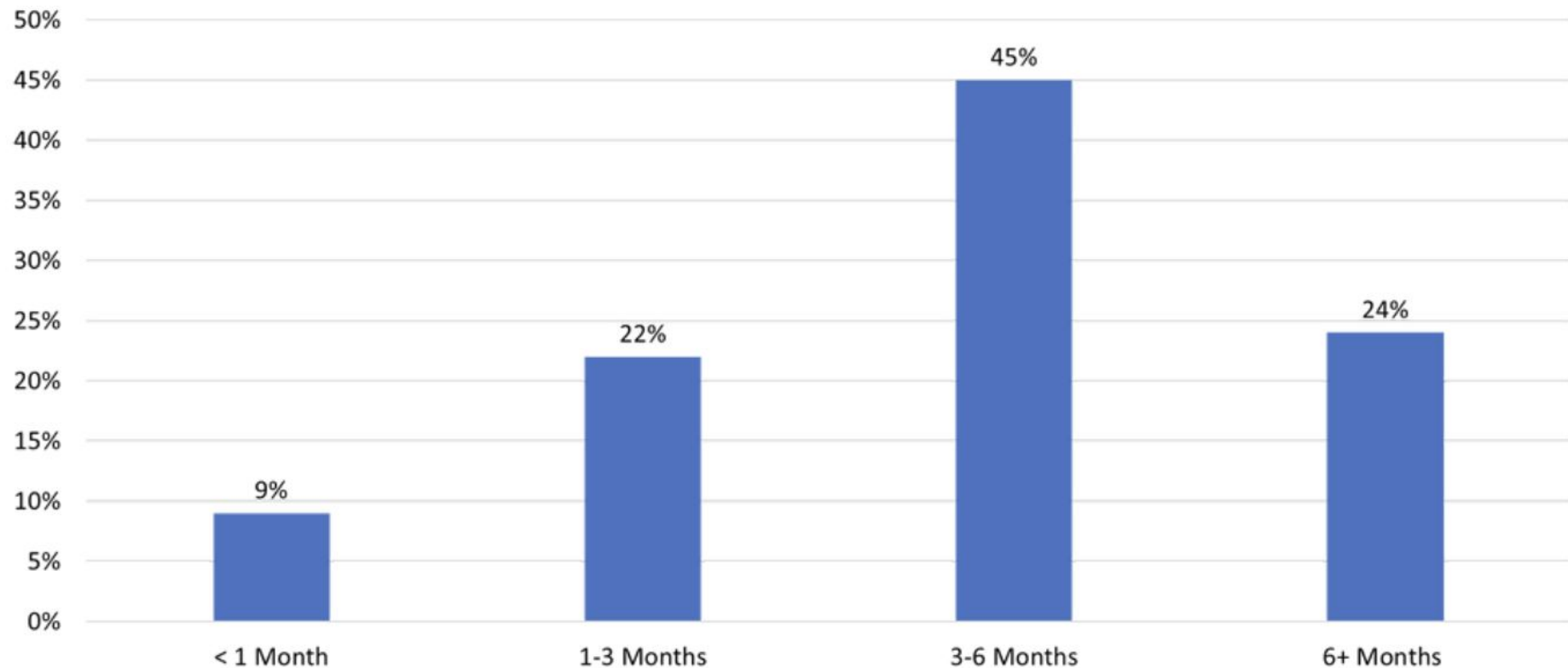
Data Science Professionals vs. AI Professionals



Data Science Professionals Residency Status Breakdown



If you are currently unemployed or were impacted by layoffs and found a new role, *how long did the job search take?*



Build A Career In Data Science (2020)

Part I: Getting Started With Data Science

- What is Data Science?
- Data Science Companies
- Getting the Skills
- Building a Portfolio

Part II: Finding Your Data Science Job

- The Search: Identifying the Right Job for You
- The Application: Resumes and Cover Letters
- The Interview: What to Expect and How to Handle It
- The Offer: Knowing What to Accept

Part III: Settling Into Data Science

- The First Months on the Job
- Making an Effective Analysis
- Deploying a model into production
- Working with Stakeholders

Part IV: Growing In Your Data Science Role

- When your Data Science Project Fails
- Joining the Data Science Community
- Leaving Your Job Gracefully
- Moving up the Ladder

1. The Search: The Right Job

Know where to start:

- LinkedIn, Indeed, and Glassdoor
- POCIT and Tech Ladies, for people of color and women in technology respectively
- job boards for specific types of companies like startups (AngelList) and technology (Dice)
- specific company's careers page
- X (Twitter), Bluesky, & Meetups

Watch for red flags:

- No description of company or job
- A job that is actually three jobs

Demand Fit, not Perfection

Advice for new grads

*If you're about to or just graduated college, **your most relevant skill is your education.** Your **data science portfolio** will be helpful here too. When you're searching for jobs, look for positions specifically titled **"New Grad," "Junior," "Associate,"** and **"Entry-level."** Also look at your career center for help and go to any job fairs that happen on campus. Internships are relevant – less for what work skills you learned and more that it shows you can come to an office each day, be professional and productive.*

On Job Descriptions

*The first thing to keep in mind is that **job descriptions are generally wish lists** with some flexibility. If you meet 60% of the requirements (e.g. you're a year short of their required work experience or haven't worked with one component of their tech stack), but are otherwise a good fit, you should still apply.*

2. The Application

- Resume *and* cover letter should be compelling
 - Resume:
 - goal is to get you an interview, not a job
 - Better be skimmable
 - Includes: contact info, education, experience, and skills

SARA JONES

San Francisco, CA · 534-241-6264

sarajones@gmail.com · linkedin.com/in/sarajones · sarajones.github.io · github.com/sarajones

EXPERIENCE

JUNE 2019 – PRESENT, SAN FRANCISCO, CA

DATA SCIENCE FELLOW, AWESOME BOOTCAMP

- Built a web application in Python that recommends the best New York City neighborhood to live in based on someone's budget, lifestyle preferences, and work
- Analyzed 2,200 New York Times business articles (obtained via API) using natural language processing (TFIDF and NMF), visualizing how topics changed over time

AUGUST 2017 – JUNE 2019, SAN FRANCISCO, CA

INVESTMENT CONSULTANT, BIGCO

- Created a forecasting model in Python that boosted quarterly revenue by 10%
- Automated generating weekly market and industry trend reports

SEPTEMBER 2016 – JUNE 2017, NEW ORLEANS, LA

INTRODUCTION TO STATISTICS TEACHING ASSISTANT, COOL UNIVERSITY

- Led weekly review sessions of sixty students, earning a 4.86/5 rating in evaluations
- Created and open-sourced study guides that have been downloaded over 1,500 times

JUNE 2016 – AUGUST 2016, NEW ORLEANS, LA

ECONOMICS RESEARCH ASSISTANT, COOL UNIVERSITY

- Conducted an in-person experiment on decision-making with 200 participants, using cluster analysis to analyze the results in Python
- Published the resulting paper in the Journal of Awesome Economics

EDUCATION

JUNE 2017, NEW ORLEANS, LA

BA ECONOMICS, STATISTICS MINOR COOL UNIVERSITY

GPA 3.65/4.0

Relevant Coursework: Linear Algebra, Introduction to Regression and Statistical Computing, Experimental Design, Econometrics, Elements of Algorithms and Computation

SKILLS

- | | |
|--------------------|----------------|
| ▪ Python | ▪ Pandas |
| ▪ SQL | ▪ Seaborn |
| ▪ Machine learning | ▪ Scikit-learn |
| ▪ Git | ▪ NumPy |

2. The Application

- Resume *and* cover letter should be compelling
 - Resume:
 - goal is to get you an interview, not a job
 - Better be skimmable
 - Includes: contact info, education, experience, and skills
 - Cover Letter
 - Should highlight both why you want *this* job and why *you* are a particularly good fit
 - Demonstrate your research/knowledge about the company and position
 - Tailor these for each job
 - Allow them to be machine-searchable
 - Referrals are a way to back-door past the algorithms
 - If contacting someone (LinkedIn, Twitter), give them a reason to read your message

SARA JONES

New York, NY · 534-241-6264

sarajones@gmail.com · linkedin.com/in/sarajones · sarajones.github.io · github.com/sarajones

GREETING

Dear Jared,

INTRODUCTORY PARAGRAPH

I am writing to express my strong interest in applying for the Data Scientist position at Awesome Company. I've enjoyed reading Awesome Company's data science blog since it started 8 months ago. The post on using topic modeling to automatically generate tags for your support articles was immensely helpful in one of my own projects to classify articles in the New York Times business section.

1-2 PARAGRAPHS OF DATA SCIENCE WORK EXAMPLES

I recently graduated from Awesome Bootcamp, a full-time, 3-month Data Science immersive. At Awesome Bootcamp, I designed, implemented, and delivered data science projects in Python involving data acquisition, data wrangling, machine learning, and data visualization. For my final project, I gathered 3,000 neighborhood reviews and ratings from Neighborhood Company. By using natural language processing on the reviews and available listings from Real Estate Company's API, I built a recommendation system that will match you to a neighborhood based on your budget, preferences, and a free-text description of your ideal neighborhood. You can try it out here: myawesomewebapp.com.

Prior to Awesome Bootcamp, I was an Investment Consultant at BigCo. When I joined, my team of six was all using Excel. While exceeding my targets, I began automating common tasks in Python, such as generating a weekly market and industry trends report, saving the team hours each week. I then developed a tailored curriculum to teach them Python. The initiative was so successful the company asked me to develop a full 2-day workshop and flew me out to three other offices to teach it, reaching over 70 consultants.

CLOSING PARAGRAPH

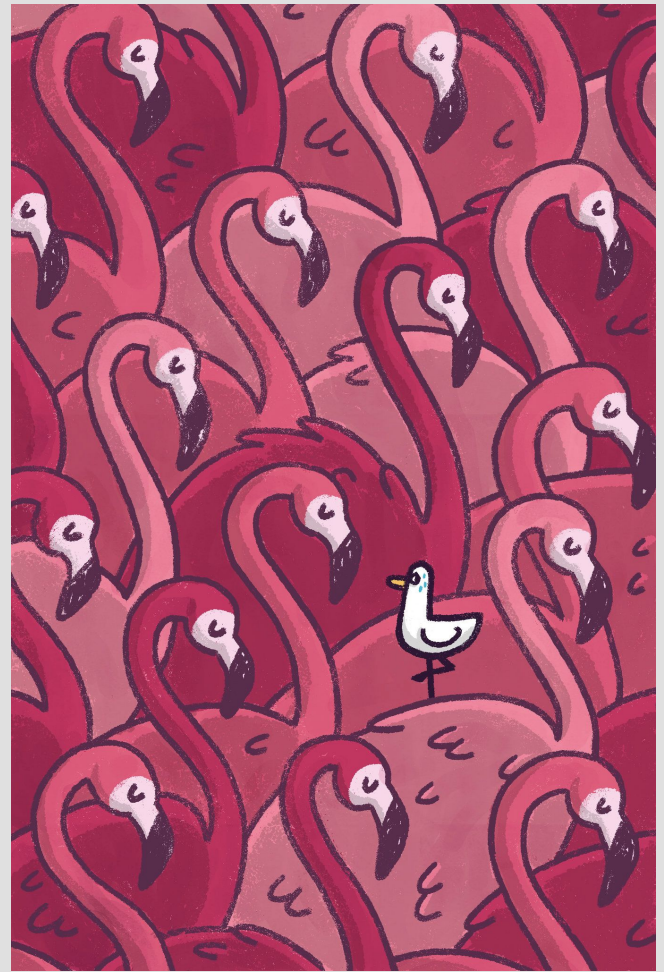
I am confident that my expertise in Python, academic training in Economics and Statistics, and experience delivering business results would make me a great fit for the Data Science team. Thank you for your consideration.

SIGNOFF

Sincerely,
Sara Jones

Figure 6.2 An example cover letter with highlights showing the different components

How are you
going to stand
out in the
crowd?



adapted from Sean Kross

3. The Interview

Basic understanding of you/position; assessment of fit

Are you able to do the job? Are you a good fit?

Take-home assignment to determine your
problem-solving and technical skills

Tie up loose ends, presentation,

	1. Phone interview
	2. In-person interview
	3. Case study
	4. Leadership interview and offer

4. The Offer

1. Offer is coming - general outline of offer coming your way
2. Company makes an offer - often in email; get it in writing; includes salary, start date,
3. You respond - thank them for offer and let them know you're excited to look it over in detail
4. You negotiate - lay out what you want/need to accept the offer; best for you
 - a. What is negotiable? Salary (5%), start date, vacation, flexibility, earlier review (earlier raise), educational benefits, budget for travel/conferences, benefits (less often), options
 - b. Best leverage: a competing offer
5. You decide - communicate final decision

Minimal Advice: Getting Your First Job in Data Science (Summary)

- Learn one programming language extraordinarily well (Python, R).
- Learn SQL well.
- Learn how to set up and interact with cloud computing services.
- Know how to *think* and *communicate* about data
- Create a resume and have a few people with relevant knowledge help you revise it.
- Establish a professional web presence.
- Be prepared to apply to many dozens of jobs.

What you all have done...

COGS 108: What we've learned

01: Data Science Python, & Version Control

02: Version Control & Data Intuition

03: Data wrangling w/ pandas & Ethics

04: Formulating Data Science ?s & Dataviz

05: Data Analysis: Descriptive & EDA

06: Inference

07: Text Analysis

08: Machine Learning

09: Geospatial Analysis

10: DS Communication & DS Jobs

COGS 108: Final Project Lessons

1. Asking the right question up front really helps
2. Finding the data you need is a skill
 - a. ...so is knowing if the data are reliable
 - b. ...and if they can answer your question
 - c.and recognizing what information you don't have
3. Data Visualization and storytelling are important skills.
4. Determining which analytical approach is best is HARD.
5. Programming is merely a piece of the puzzle for data scientists.

...so where are we going?

- **Data aren't going anywhere** | continued utilization (exploitation?) of data across products, apps, & platforms; remember: if you don't pay for the product, you and your data are the payment
- **(continued) Rapid development in AI & ML** | continued expansion beyond text, decrease in hallucinations, ~~ability to cite sources well~~, hopefully increased interpretability and importance placed on ethics
- **Automation beyond now** | self-driving cars, more targeted ads, self-serve all the things
- **Increased regulation** | I don't see it staying how it is much longer, but Congress is slow

“A lot of times, people are talking about bias in the sense of equalizing performance across groups. They’re not thinking about the underlying foundation, whether a task should exist in the first place, who creates it, who will deploy it on which population, who owns the data, and how is it used?”

-Timnit Gebru

COGS 108 Thank yous!

Reuben Chatterjee

Yueyan Tang

Liz Kaplan

Divya Prabhu

Yueying Dong

Tori Hennessy

Jonathan Ahern

Tyler Chang

Ashley Chavarria

Coco Luo

Shrieyaa Jayanthi

Will Kim

Sehee Kim

Ricky Zhu

Jorge Ramos

Nick Liu

All of you for your patience,
feedback, and time!

You all are the future of data science!

So, if you remember anything from this course...



Ethics should always be a priority in your work.



Data wrangling is a puzzle and a big part of the job. When done well, it's not boring!



Data science is a competitive, but rewarding field. You have a chance to make a big difference!



Your grade in this course is probably not predictive of future success.



**My hope is that all of you go on to
(continue to) be good people who are
happy, balanced, curious, engaged with
the world, fulfilled & successful.**

Thanks for taking COGS 108!
